COLLEGE OF INFORMATION SCIENCE AND ELECTRONIC ENGINEERING

(ISEE)

Electronic Science and Technology Curriculum (2020)

(Updated in May 2023)

1. Program Educational Objectives

Electronic Science and Technology focuses on the design and application of large and small scale electronic systems to process and transmit energy and information in this professionally accredited degree. This program has a balanced core and a flexible elective structure designed to equip our graduates in the areas of electromagnetics, electronics and microelectronics, integrated circuits, digital signal processing and photonics, etc. We expect our graduates to progress to positions in industry or graduate study, or to carry the attributes of an engineering education into other disciplines. These programs of study must include mathematics and basic sciences, fundamentals and applications in several engineering sciences, and team-based experience in the process of design, where theory is applied in the context of real-world needs and constraints, and where judgement must be exercised. Our graduates should be able to think critically in solving problems and managing tasks and communicates effectively in multidisciplinary professional environments. To be a responsible member of the engineering profession, each graduate must be aware of social, ethical, environmental and economic factors and constraints on engineering activity, and must understand the importance of these matters in a global context. We aspire to have our graduates exhibit intellectual depth and creativity, uphold high ethical standards, and show a commitment to the betterment of society through service and professional work.

2. Graduation Requirement

The curriculum leading to the degree of Bachelor of Electronic Science and Technology from ZJU requires **135** credits and is organized into required courses, technical electives, liberal education, Chinese language and society study for international students, and other electives.

3. Required Courses

3.1 Orientation and Professional Development

These courses introduce the opportunities and resources our institute and curriculum can offer you as you work to achieve your career goals. They also provide the skills to work effectively and successfully in the engineering profession.

Course Code	Course Name	Credit
ENGR 100	Engineering Orientation	**1**

3.2 Foundational Mathematics and Science

These courses stress the fundamental mathematical and scientific principles upon which the engineering discipline is based.

Course Code	Course Name	Credit
MATH 221	Calculus I	4
MATH 231	Calculus Ⅱ	3
MATH 241	Calculus III	4
MATH 286	Intro Differential Equations	4
PHYS 211	University Physics: Mechanics	4
PHYS 212	University Physics: Elec& Mag	4
PHYS 213	Univ Physics: Thermal Physics	2
	Total required	**25**

3.3 Technical Core

These courses stress fundamental concepts and basic laboratory techniques that comprise the common intellectual understanding of electronic science and technology.

Course Code	Course Name	Credit
ECE 110	Intro to Electronics	3
ECE 210	Analog Signal Processing	4
ECE 313	Probability with Eng App	3
ECE 329	Fields and Waves I	3
ECE 340	Semiconductor Electronics	3
ECE 342	Electronic Circuits	3
ECE 343	Electronic Circuits Laboratory	1
ECE 385	Digital Systems Laboratory	3
ECE 498FP	Fundamentals of Optoelectronics	3
ECE 498COD	Computer Organization and Design	3.5
ECE 498SS	Signals and Systems	4
ECE 498PI	Project Internship	2
ECE 498FTP	Final Term Project	12
	Total required	**47.5**

4. Elective Courses

4.1 Technical Electives

This elective requirement gives each student the freedom to develop further in mathematics, science and a technical course of study in Electronic Science and Technology of considerable breadth and focus. Students must complete a minimum of **14.5** credits on technical electives.

Course Code	Course Name	Credit
CS 101	Intro to Computing: Engrg & Sci	3
CS 225	Data Structures	4
MATH213	Introduction to Discrete Mathematics	3

MATH257	Linear Algebra with Computational Applications	3
CHEM102	General Chemistry Ⅰ	3
CHEM103	General Chemistry Lab Ⅰ	1
CHEM104	General Chemistry II	3
CHEM105	General Chemistry Lab II	1
ECE 120	Introduction to Computing	4
ECE 310	Digital Signal Processing	3
ECE 311	Digital Signal Processing Lab	1
ECE 314	Probability in Engineering Lab	1
ECE 330	Power Circuits & Electromechanics	3
ECE 350	Fields and Waves II	3
ECE 365	Data Science and Engineering	3
ECE 374	Introduction to Algorithms & Models of Computation	4

4.2 Liberal Education

The liberal education courses develop students' understanding of human culture and society, build skills of inquiry and critical thinking, and lay a foundation for civic engagement and lifelong learning. To satisfy the liberation education requirements, students must take all courses for grade and complete courses based on the table below:

Category	**Credit**
1) Humanities & Arts (2 courses)	6
2) Social & Behavioral Sciences (2 courses)	6
3) Culture Studies	6
Western/Comparative Culture(s) (1 course)	
Non-Western Culture(s) (1 course)	
Total required	**18**

Appropriate choices in the Social and Behavioral Sciences and in the Humanities and Arts will ensure that these courses also fulfil the requirements in the areas of Western and Non-Western Cultures.

4.3 Free Electives

These unrestricted electives give the student the opportunity to explore any intellectual area of unique interest. This freedom plays a critical role in helping students to define research specialties. Students are encouraged to take cross-discipline courses as free electives for **9** credits.

5. Chinese Language and Society

These courses are compulsory for all international students. They introduce Chinese modern history, social development, government policies, etc., and help students improve their Chinese and understanding about China.

International students need to complete five courses (20 credit hours) in Chinese language study and Chinese society to fulfill the graduate requirement.

Course Code	**Course Name**	**Credit**
CHIN 1001	Chinese Ⅰ	4
CHIN 1002	Chinese Ⅱ	5
CHIN 1003	Chinese Ⅲ	4
CHIN 1004	Chinese Ⅳ	4
CULT 2001	China Survey	3
	Total required	**20**

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MATH 285	Intro Differential Equations	3
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ECE 313	Probability with Eng App	3
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ECE 340	Semiconductor Electronics	3
ECE 385	Digital Systems Laboratory	3
ECE 498FP	Fundamentals of Optoelectronics	3
ECE 498COD	Computer Organization and Design	3.5
ECE 498SS	Signals and Systems	4
ECE 498PI	Project Internship	2
ECE 498FTP	Final Term Project	12
	Total required	**47.5**

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This elective requirement gives each student the freedom to develop further in mathematics, science and a technical course of study in Electronic Science and Technology of considerable breadth and focus. Students must complete a minimum of **15.5** credits on technical electives.

Course Code	Course Name	Credit
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These courses are compulsory for all international students. They introduce Chinese modern history, social development, government policies, etc., and help students improve their Chinese and understanding about China.

International students need to complete courses in Chinese language study and Chinese society to fulfill the graduate requirement.

		2020-2022	**2023-**
Course Code	**Course Name**	**Credit**	**Credit**
CHIN 1001	Chinese Ⅰ	4	4
CHIN 1002	Chinese Ⅱ	5	4
CHIN 1003	Chinese Ⅲ	4	4
CHIN 1004	Chinese Ⅳ	4	4
CHIN1005	International Chinese Language Proficiency and Testing		2
CULT 2001	China Survey	3	3
	Total required	**20**	**21**